Finite Lie-group witnesses for chiral polyhedra

Human Compression project

1 The finite source obligations

Leemans and Liebeck [1] prove that every nonabelian finite simple group except $A_7, \mathrm{PSL}_2(q), \mathrm{PSL}_3(q)$ and $\mathrm{PSU}_3(q)$ admits a chiral polyhedron. We recover the existence assertions for the thirty-five parameter entries in Table 1. These are existing results, not new existence theorems.

There are twenty-one simple entries in their list (4.1), physical page 7:

$$L_4^\epsilon(2), L_4^\epsilon(3), L_4^\epsilon(4), L_5^\epsilon(2), L_6^\epsilon(2),$$
$$\mathrm{PSp}_4(3), \mathrm{Sp}_4(4), \mathrm{Sp}_6(2), \mathrm{PSp}_6(3), \mathrm{Sp}_8(2),$$
$$\Omega_7(3), \Omega_8^\epsilon(2), \mathrm{P}\,\Omega_8^+(3), \Omega_{10}^\epsilon(2).$$

Both signs occur, with $L^+ = \mathrm{PSL}$ and $L^- = \mathrm{PSU}$. The additional printed $\mathrm{Sp}_4(2) \cong S_6$ is not simple; it is not a positive simple-case obligation. The eleven remaining finite exceptions in Section 4(2), physical page 8, are

$$L_5^\epsilon(3), L_6^\epsilon(3), L_8^\epsilon(2), \mathrm{Sp}_6(4), \mathrm{PSp}_8(3), \mathrm{Sp}_8(4), \Omega_{12}^\epsilon(2).$$

The last three entries are $G_2(3), G_2(4), {}^2F_4(2)'$, treated at the end of Section 3, physical page 6. We do not include the nonsimple $G_2(2)$, nor turn its derived group $\mathrm{PSU}_3(3)$ or ${}^2G_2(3)' \cong \mathrm{PSL}_2(8)$ into positive exceptions.

Lemma 1. *Let $G = \langle x, t\rangle$, where $|t| = 2$. If $|w(x,t)| \neq |w(x^{-1},t)|$ for a group word w, no automorphism of G inverts x and fixes t.*

Proof. Such an automorphism would carry the first word value to the second. It preserves element order, which contradicts the inequality. □

This is the elementary word criterion already proved in the accepted twelve-sporadic component [4]; it is restated to make the present argument self-contained. In every row we take $w(x,t) = xtx^2tx^3t$.

Theorem 1. *Each parameter entry in Table 1 has generators x, t satisfying conditions (i)–(iii) of [1, Theorem 1.1]. It therefore admits a chiral polyhedron of the displayed type $\{|x|, |x^{-1}t|\}$, with that named simple group as full automorphism group.*

Proof. The exact checks described below establish membership, $|t| = 2$, full generation, and the unequal word orders. Lemma 1 rules out even noninvolutory reversing automorphisms. Put $y = x^{-1}t$. Then $xy = t$. An element of $\langle x\rangle \cap \langle y\rangle$ commutes with both x, y, so is central. The named groups are nonabelian simple, hence this intersection is trivial. The first two orders exceed two. The retained Schulte–Weiss reconstruction theorem [2], in the precise form used in [1, Section 1], now gives the stated geometric conclusion. □

2 Named models and exact verification

Every `pair.g` is a literal permutation record containing the two witnesses, the full ambient generating list, the parameters, the degree and the expected orders. A fresh GAP process resets the same recorded constructor seed and calls `SimpleGroup` with those parameters. It requires exact equality of the saved and newly constructed generating lists. There is no recognition of a group from its order.

The named-model identifications are retained GAP 4.15.1 construction and catalogue premises [3]. The `L`, `U` and `S` branches construct `PSL`, `PSU` and `PSp`. Their projective matrix constructions and subsequent faithful smaller actions are the documented named constructors. The odd and signed even orthogonal branches take the derived subgroup of the appropriate special orthogonal action on lines, followed when applicable by the constructor's smaller action. Their identification and simplicity are retained; neither the stored `Size` nor a simplicity flag proves the identification here. The three exceptional calls are explicitly

$G_2(3)$	`PrimitiveGroup(351,7)`
$G_2(4)$	`PrimitiveGroup(416,7)`
${}^2F_4(2)'$	`PrimitiveGroup(1600,20).`

The exact primitive catalogue records name these groups. The constructor, classical projective interface, primitive loader and actual catalogue files are pinned in the dependency ledger. Correctness of GAP's exact permutation algorithms and the named library inputs is explicit computational infrastructure, not a new classification proof.

For every row the checker reconstructs H from the saved ambient generators without supplying its order. It verifies $H = G$, computes $|H|$, checks $x, t \in H$ and $|t| = 2$, and forms $K = \langle x, t \rangle$ without supplying its order. It verifies $|K| = |H|$ and $K = H$. Membership and equal finite order already imply generation; the latter equality is a redundant explicit check. The four table entries are then computed by exact permutation operations. The new groups are checked to have no preassigned size before the relevant calculations.

A separate standard-library Python parser reads the original cycle notation directly, checks disjoint cycles, and composes the corresponding right-action permutations. Cycle lengths and integer least common multiples reproduce all four orders and $|t| = 2$, independently of GAP's word calculations. It also parses every ambient generator, but it does not purport to duplicate GAP's large subgroup computations.

The source roster is constructed independently as the explicit sets of parameters above. Missing, duplicate, nonsimple, misrouted and displaced entries are rejected. Seven GAP controls damage parameters, ambient generators, the involution, membership, generation, an order, and the word inequality. All must reject before producing a positive certificate. Both Python entry points reject optimized execution before output work. Every replay is finite; a timeout or nonclean terminal result grants no certificate. The recorded searches merely supplied candidates and are not run by verification.

3 Attribution and limits

The root agent supplied the bounded candidate searches and the acquired word argument; this component's author supplies the deterministic certificate, independent literal oracle, source routing and explanation. All original candidate and execution bytes are retained. An initial author setup used a directory ordinal instead of the actual recorded seed for some constructors; that failed preliminary setup is preserved outside the scientific replay. The final replay uses the recorded seed and changes no candidate.

These are thirty-five parameter entries, not thirty-five newly classified groups. In particular $L_4(2) \cong A_8$ overlaps the alternating family, and $U_4(2) \cong \mathrm{PSp}_4(3)$ occurs twice under the source's parameters. The group identities, geometric reconstruction and original existence theorem retain their attribution. This component does not settle the infinite Lie families, ancillary enumerations, or the whole original source, and adds no measurement or conjecture claim.

4 The complete finite table

G	$\lvert x\rvert$	$\lvert x^{-1}t\rvert$	$\lvert w\rvert$	$\lvert w^{-}\rvert$
$L_4(3)$	8	5	9	12
$U_4(3)$	8	9	7	5
$L_4(4)$	30	85	30	15
$U_4(4)$	65	30	65	15
$L_5(2)$	15	15	3	7
$U_5(2)$	12	18	12	11
$L_4(2)$	15	6	5	6
$U_4(2)$	9	12	6	5
$L_6(2)$	12	31	28	31
$U_6(2)$	6	18	10	11
$S_4(3)$	4	9	4	5
$S_4(4)$	15	5	17	10
$S_6(2)$	6	9	12	10
$S_6(3)$	13	9	24	9
$S_8(2)$	30	30	8	6
$O_7(3)$	13	13	7	8
$O_8^+(2)$	12	10	10	12
$O_8^-(2)$	30	6	7	6
$O_8^+(3)$	15	12	15	20
$O_{10}^+(2)$	12	31	12	15
$O_{10}^-(2)$	11	30	6	15
$L_5(3)$	121	9	18	121
$U_5(3)$	24	28	21	40
$L_6(3)$	52	121	10	39
$U_6(3)$	91	80	13	56
$L_8(2)$	30	105	62	124
$U_8(2)$	85	66	12	85
$S_6(4)$	51	63	20	63
$S_8(3)$	13	84	24	40
$S_8(4)$	195	126	255	65
$O_{12}^+(2)$	63	18	24	18
$O_{12}^-(2)$	24	102	34	60
$G_2(3)$	13	12	7	6
$G_2(4)$	7	13	6	15
${}^2F_4(2)'$	12	8	13	12

Table 1: Exact orders in the saved witnesses. $L_n(q), U_n(q), S_n(q)$ mean $\mathrm{PSL}_n(q), \mathrm{PSU}_n(q), \mathrm{PSp}_n(q)$; $O_n^{\pm}(q)$ means $\mathrm{P}\,\Omega_n^{\pm}(q)$ and $O_7(3) = \Omega_7(3)$. Let $w = xtx^2tx^3t$ and $w^- = w(x^{-1}, t)$.

References

[1] D. Leemans and M. W. Liebeck, *Chiral polyhedra and finite simple groups*, fixed arXiv:1603.07713v3, Sections 1, 3 and 4.

[2] E. Schulte and A. Ivić Weiss, *Chiral polytopes*, Applied Geometry and Discrete Mathematics, DIMACS Series 4 (1991), 493–516.

[3] The GAP Group, *GAP—Groups, Algorithms, and Programming*, version 4.15.1, reference manual Sections 39.15-14 and 50.2; PrimGrp 4.0.1. Exact local implementation and data pins accompany this proof.

[4] Human Compression project, *Chirality detected by the order of one word*, accepted twelve-sporadic component, final edition v2; its distinct review and root adoption are retained.